



Learning from data for physicists

Bayesian and machine learning methods

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<https://go.osu.edu/lfd>

This text is aimed at physicists seeking to learn from data using Bayesian methods. It is designed primarily for use in an advanced-level course but should also be broadly accessible for individual study to those with a physics background.

The book format is powered by [Jupyter Book](#). The [complete book as pdf](#) (>900 pages) is also available.

Major parts: Bayesian methods for scientific modeling; Advanced Bayesian methods; MCMC sampling; Machine learning basics; Probabilistic machine learning

Features:

- Demonstration code and notebooks are integrated in the Jupyter Book.
- Four types of problems: checkpoints (dropdown hints and answers); exercises (answers at end of sections); guided problems (ipynb); projects (complex problems).
- Interactive widgets: pdf explorer; Bayesian coin tossing; credible intervals; bivariate gaussians; gaussian processes; MCMC simulations; and more being added!